

TECHNICAL ROADMAP



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Naming Convention: [Track]_[University]_[TeamName]_Roadmap.pdf

TEAM INFORMATION

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SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

Climate Resilience Copilot (AI-Native Heatwave Electrical-Fire Early Warning for ASEAN Buildings) addresses a concrete climate-adaptation challenge in ASEAN cities: extreme heatwaves increase cooling demand in buildings, dormitories, campuses, and dense urban zones, which can overload electrical systems and create pre-fire conditions before smoke or flame appears. Instead of reacting only after visible danger emerges, the solution detects climate-triggered compound risks earlier by monitoring high temperature, abnormal power load, and humidity patterns that may indicate wiring stress, short-circuit risk, or unsafe electrical operation. By transforming distributed IoT telemetry into real-time digital-twin context, the system helps operators intervene earlier, reduce heatwave-related infrastructure risks, and strengthen urban resilience.

The solution combines Matter and Zigbee IoT telemetry, FIWARE Orion digital twins, and an MCP-based AI orchestration layer. Sensor readings are normalized into NGSI-v2 entities, evaluated by a rule-first risk engine, and enriched by AI reasoning for human-readable rationale and recommended actions. The key AI/MCP contribution is a tool-mediated resilience agent that observes live digital-twin context, invokes risk-computation tools, explains compound risk conditions, recommends response playbooks, and coordinates auditable alert/action workflows. The project is therefore not a conventional fire-alarm system; it is a climate-resilience decision-support platform for detecting early infrastructure stress caused by heatwave-driven energy demand.

When warning or critical thresholds are detected, the system publishes alerts, visualizes risk on a dashboard, and can invoke controlled response actions such as alert signaling or operator-approved simulated power isolation with ACK/ERROR tracking. The current MVP uses a simulated command path that can later connect to real hardware bridges. The demo targets ≤ 5 -second detection, zero false negatives in predefined critical scenarios, and scalable deployment from one room or building to multi-zone ASEAN communities exposed to climate-driven heat stress.

SECTION 2: TECHNICAL ARCHITECTURE

2.1 System Components:

Inputs: The system receives real-time telemetry from Matter emulators and optional Zigbee devices, including temperature ($^{\circ}\text{C}$), active power (W), humidity (%RH), device status (onOff), timestamps, and zone context. It can also receive operator requests through the dashboard and Telegram/OpenClaw gateway for status queries, manual evaluation, and controlled commands.

Processing Core: The AI contribution is implemented as an MCP-based resilience agent running in the MCP Agent layer (Node.js, port 3002). It follows an **Observe** $\rightarrow$ **Assess** $\rightarrow$ **Explain** $\rightarrow$ **Act** $\rightarrow$ **Audit**

workflow. First, it observes real-time NGS-v2 digital-twin entities from FIWARE Orion. Second, deterministic tools compute baseline risk using transparent safety rules. Third, the AI reasoner interprets compound climate-risk signals and generates human-readable explanations, response recommendations, and operator guidance. Fourth, the MCP Agent invokes controlled alert and command tools under policy guardrails and human-in-the-loop constraints for sensitive actions. Finally, every alert and command result is recorded with ACK/ERROR tracking for accountability. Context is stored centrally in FIWARE Orion (NGSI-v2), while FIMAT Agent and Zigbee Bridge normalize telemetry into digital twins. Proxy and dashboard services expose the live state for human monitoring.

Outputs: The system delivers risk scores, risk levels, natural-language rationale, dashboards, alert events, command executions with ACK/ERROR tracking, and optional multi-channel notifications via the OpenClaw Telegram gateway. In the current MVP baseline, ACK/ERROR results are generated by the command-executor simulation unless a real hardware bridge is connected.

2.2 Architecture Diagram:

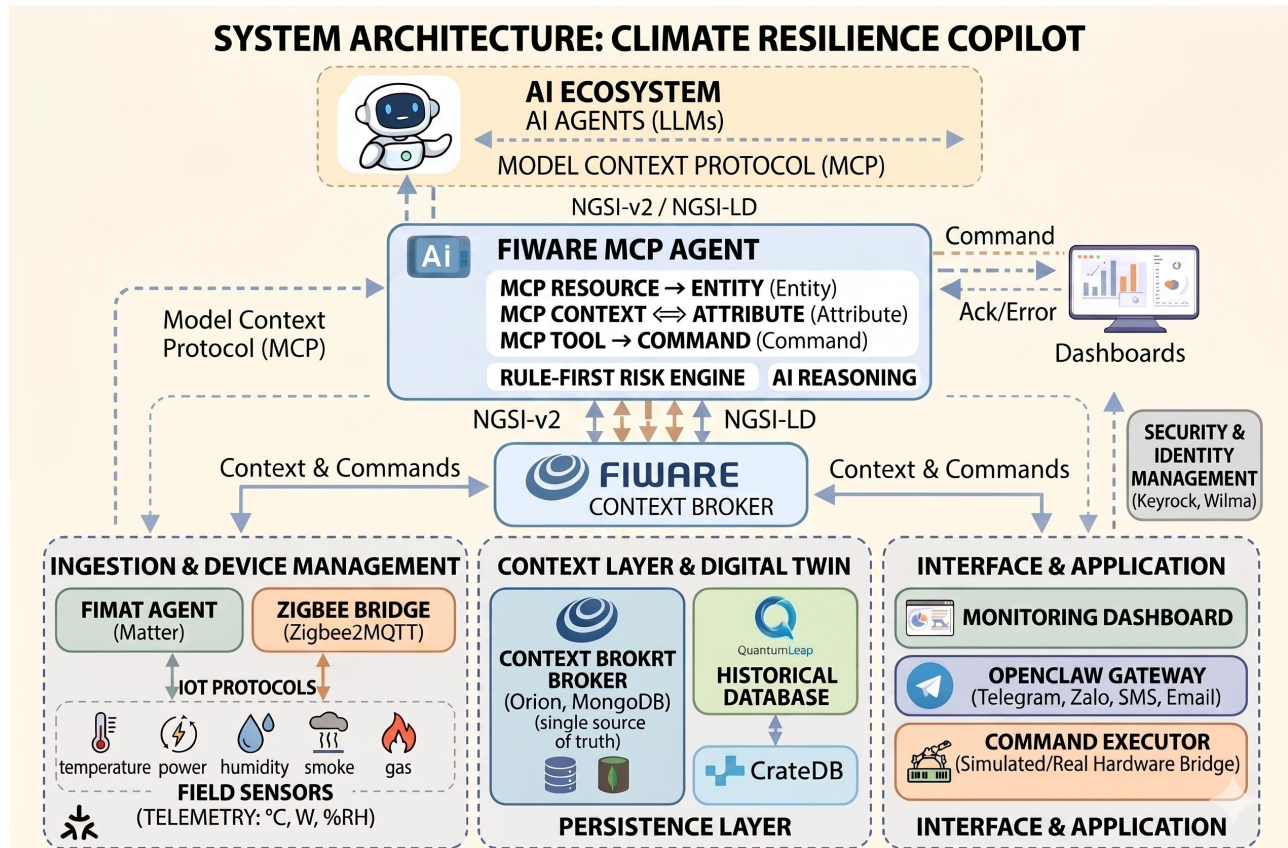


Figure 1. Reference architecture concept for integrating AI Agents (LLMs), MCP, FIWARE Context Broker, IoT Agents, applications, and dashboards. In this project, the same principle is applied through FIMAT Agent / Zigbee Bridge -> Orion -> MCP Agent -> Dashboard / Action / Notification.

As illustrated in Fig. 1, in the climate-resilience scenario, this pipeline converts heatwave-related telemetry such as room temperature, active power, and humidity into a digital-twin risk state, allowing the MCP Agent to detect pre-fire infrastructure stress and coordinate preventive response. The MCP Agent implements an agentic workflow: Observe digital-twin context from Orion, Assess risk through deterministic tools, Explain the situation using AI reasoning, Act through controlled command tools, and Audit all outcomes through ACK/ERROR records.

Architecture details derived from the source code and docs:

- Devices and Emulators: matter-emulators simulate TemperatureSensor, HumiditySensor, and SmartPlug telemetry.
- Ingestion Layer: FIMAT Agent (port 3000) converts Matter events to NGSI-v2 and upserts entities into Orion. Zigbee Bridge (port 3003) performs the same role for zigbee2mqtt and MQTT devices.
- Context Layer: FIWARE Orion Context Broker (port 1026, backed by MongoDB 4.4) is the single source of truth for digital-twin state.
- Decision Layer: MCP Agent (port 3002) polls Orion every 5 seconds, computes risk, enriches explanations with AI, publishes AlertEvent and ZoneRisk entities, and recommends or invokes policy-gated protective commands when policy allows, with sensitive actions remaining operator-approved.

- Visualization Layer: monitor-dashboard (port 8080) displays devices, alerts, maps, simulator controls, and action visibility through a CORS proxy (port 3001).
- Notification Layer: OpenClaw Gateway (port 3004) provides Telegram-based routing, alert push, and operational skill shortcuts.

Core non-negotiable system flow: Device/Event -> FIMAT -> Orion -> Dashboard/MCP -> Action/Ack.

SECTION 3: AI APPROACH & MODEL SELECTION

Primary AI Approach: ☐ Machine Learning ☐ NLP ☐ Computer Vision ☒ Generative AI ☐ Other: Rule-based risk engine and digital twin orchestration

AI/MCP Contribution: The project uses MCP as a controlled tool-use layer between AI reasoning and the FIWARE digital-twin environment. Instead of allowing the AI model to directly control devices, the MCP Agent exposes a limited set of auditable tools for querying context, computing risk, publishing alerts, invoking commands, and answering operator questions. This design makes the AI contribution practical, explainable, and safe for climate-resilience operations.

MCP Capability	Implemented Function	Contribution to Climate Resilience
Query context	/tools/query_entities	Reads Orion entities
Compute risk	/tools/compute_risk, /risk	Warning/critical score
Explain	/tools/ai_chat	Human-readable rationale
Alert	/tools/publish_alert	AlertEvent
Act/Audit	/tools/invoke_command	Simulated/operator-approved action with ACK/ERROR log

Model Selection: The MCP Agent is designed to work with an optional LLM endpoint through AI_ENDPOINT, AI_API_KEY, and AI_MODEL. The current configuration can use a lightweight generative AI model when available, while the core safety path remains rule-first. This hybrid strategy is intentional: deterministic rules provide low-latency and fail-safe detection, while generative AI improves the quality of explanation, operator guidance, scenario interpretation, and response playbook generation.

Reasoning: In a life-safety climate-resilience scenario, AI should not become a single point of failure. Therefore, critical detection is performed by transparent thresholds and compound-risk rules. The AI layer contributes by interpreting multi-sensor context, explaining why a risk level was assigned, translating technical telemetry into actionable language, recommending next steps, and supporting operator Q&A through MCP tools. If the AI endpoint is unavailable, the system still detects risk and publishes alerts. If AI is available, the system becomes more understandable, interactive, and decision-supportive without sacrificing deterministic safety.

- Implemented warning thresholds: temperature $\geq 40^{\circ}\text{C}$, activePower $\geq 800\text{W}$, humidity $\geq 75\%$.
- Implemented critical thresholds: temperature $\geq 50^{\circ}\text{C}$, activePower $\geq 950\text{W}$, humidity $\geq 90\%$ combined with high power.
- Critical recommended actions defined in the response policy include REQUEST_OPERATOR_APPROVAL, SIMULATE_POWER_ISOLATION, TURN_ON_ALERT_LAMP, NOTIFY_OPERATOR, and RECOMMEND_EVACUATION_GUIDANCE. In the current MVP, these actions represent an approval-gated response policy, while hardware execution remains simulated unless a validated hardware bridge and operator authorization are available.
- Implemented MCP endpoints include /risk, /risk/all, /tools/query_entities, /tools/compute_risk, /tools/publish_alert, /tools/invoke_command, /alerts, /commands, /evaluate, and /tools/ai_chat.

SECTION 4: DATA STRATEGY & ETHICS

(Based on Workshop 2: Technical Architecture and Data Ethics)

Data Sources: The MVP uses three primary data sources: (1) synthetic telemetry from Matter emulators in the repository, (2) optional live Zigbee devices via zigbee2mqtt/MQTT, and (3) optional WiFi/Tuya integration described in docs/IOT_DEVICE_GUIDE.md. All data is operational telemetry only: temperature, humidity, power, switch status, timestamps, zone, and device identifiers.

Data Quality and Cleaning: Data quality is enforced through normalized NGSI-v2 entities, stable entity IDs, per-attribute timestamp metadata, unit metadata, serialized per-entity event processing in FIMAT Agent,

idempotent upserts, and append fallback when attributes do not yet exist. The build and test gate also requires smoke tests, a demo scenario, dashboard availability, and validation that SmartPlug contains both onOff and activePower attributes.

Licensing and Legality: The software stack relies on open technologies and packages already declared in the repository, including Node.js packages, FIWARE Orion, MongoDB images, MQTT tooling, and Telegram bot libraries. The roadmap avoids any restricted private datasets by default because the demo can run entirely on synthetic and self-generated telemetry. This keeps the hackathon setup simple and legally safe.

Bias Mitigation and Fairness: This solution minimizes classic demographic bias because it does not process personal identity data, faces, voice, or sensitive personal records. The MVP explicitly avoids PII and uses only device and zone telemetry. Fairness risks instead come from infrastructure inequality, sensor placement, and uneven environmental exposure across buildings or regions. To mitigate that, the system uses transparent rules, explicit thresholds, explainable rationale, human-in-the-loop for sensitive actions, and audit trails for every command decision.

- No PII in MVP; only telemetry device and zone context is used.
- Rule-first safety design reduces opaque AI-only decision making.
- Audit trail is mandatory for actions: who, when, why, and result.

Sensitive actions require human oversight when deployed beyond controlled demo scope.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

(Based on Workshop 3: Agile Prototyping)

The roadmap below is aligned to the actual repository status and the hackathon objective of preparing a stable semi-final prototype.

Phase	Activity / Task	Tools Used	Expected Outcome
Sprint 1 (Completed / Current Baseline)	Establish FIWARE foundation: Docker Compose for Orion and MongoDB, FIMAT Matter-to-NGSI ingestion, proxy server, dashboard, smoke scripts, entity model, and API docs.	Docker Compose, Node.js, Express, FIWARE Orion, MongoDB, PowerShell	Working end-to-end MVP baseline with live entities in Orion and dashboard visibility.
Sprint 2	Harden risk detection: refine MCP risk-engine thresholds, validate critical heat-wave and moisture-short-circuit scenarios, enforce alert dedupe and cooldown, improve rationale consistency. Improve AI/MCP reasoning templates for risk explanation, scenario classification, and recommended response playbooks.	MCP Agent, rule engine, demo-scenario.ps1, smoke-test.ps1, Orion entities	Reliable warning and critical detection within <= 5 seconds and zero false negative in critical demo paths.
Sprint 3	Strengthen action and operator workflow: command ACK/ERROR tracking, alert lamp control, simulated smart-plug response, operator approval for sensitive actions, OpenClaw Telegram integration, and clearer action visibility in dashboard. Demonstrate MCP tool-use flow where the agent queries Orion context, computes risk, publishes an alert, invokes a policy-gated simulated command or operator-approved action, and records ACK/ERROR for audit.	Command executor, OpenClaw Gateway, Telegram Bot API, dashboard modules	Demonstrable closed-loop response: detect → alert → act → acknowledge → audit.
Sprint 4	Production-readiness for hackathon judging: refine docs, FAQ, KPI scorecard, test report, architecture narrative, and replayable 3-5 minute demo.	README_HACKATHON, ARCHITECTURE.md, KPI_SCORECARD.md, DEMO_SCRIPT.md, TEST_REPORT.md	Judge-ready prototype with a replayable 3–5 minute demo, KPI scorecard, timestamped test evidence, MCP tool-use logs, and measurable climate-resilience impact metrics.

Semi-final Demo KPI Scorecard: To make the prototype measurable for judging, the team will evaluate the semi-final demo using the following KPI scorecard. The metrics focus on climate-risk detection, MCP-based AI coordination, closed-loop response, privacy, and reproducibility. These KPIs are designed for the controlled hackathon demo environment and will be reported through dashboard screenshots, API logs, command ACK/ERROR records, and a short test report.

KPI	Target
Detection latency	≤ 5 seconds
Critical false negatives	0 in predefined critical demo scenarios
Demo scenarios	Normal / warning / critical
MCP tool-use	Query → compute risk → publish alert → request/approve action
Closed-loop response	Detect → alert → recommend/request action → simulate/approve → ACK/ERROR → audit
Audit completeness	100% alert/action records include timestamp, risk level, rationale, and ACK/ERROR result.
Demo repeatability	3 scenarios replayable from scripts with timestamped evidence
Data privacy	No PII; telemetry-only MVP

Demo Storyline: The demo simulates an ASEAN campus room during an extreme heatwave. A smart plug and environmental sensors first report normal conditions, then rising temperature and power load trigger a warning state. In the critical scenario, high temperature, high active power, and humidity indicate potential electrical stress. The MCP Agent queries FIWARE Orion, computes risk, explains the cause, publishes an alert, recommends protective action, requests operator approval for sensitive actions, invokes a simulated response such as alert signaling or power-isolation simulation, and records ACK/ERROR for audit.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

Scaling Potential: The architecture already separates ingestion, context, decision, visualization, and notification layers. This makes it easy to add more buildings, campuses, or districts as new zones without rewriting the whole system. FIWARE Orion provides a unified digital-twin interface, while the MCP Agent can reason over multi-zone context consistently. The same pattern can extend from one campus in Vietnam to public housing, industrial parks, universities, and municipal pilot deployments across ASEAN countries facing heat-wave stress and electrical fire risk. More specifically, this use case is particularly well-suited to ASEAN communities where heatwaves, aging electrical infrastructure, dense housing, and growing cooling demand increasingly interact. The solution can be piloted first in university dormitories and campus buildings, then extended to schools, public housing, worker dormitories, hospitals, small businesses, and industrial parks. By focusing on low-cost telemetry, open digital-twin standards, and explainable AI-assisted response, the system provides a practical adaptation pathway for cities that need affordable climate-risk monitoring without deploying expensive centralized infrastructure.

Technical Constraints: The current prototype is still an MVP. It uses emulators for the default demo, lacks full production security and HA, depends on polling rather than full event subscriptions, and has limited long-term historical storage and analytics. AI integration is optional and environment-driven, which is good for resilience but means explanation quality depends on external model availability if enabled. Hardware reliability, MQTT infrastructure, and regional device standards must be validated separately in field deployment.

- Regional deployment path: 1 room → 1 building → 1 campus → 1 district → multi-city network.
- Multi-protocol support already planned: Matter, Zigbee, optional Tuya and WiFi integration.
- Clear extension path for multi-channel alerts: Telegram, Zalo, SMS, and Email.
- Future resilience features: historical analytics, replay timeline, CI smoke tests, richer policy control, and additional sensors such as smoke, gas, motion, and water leak.

INSTRUCTIONS FOR SUBMISSION:

1. **Strictly PDF:** Convert your final document to PDF.
2. **Visuals Matter:** Use clear diagrams for Section 2.
3. **Submission Link:** Team Leaders must upload this file to the **Official Submission Folder** [SUBMISSION - TECHNICAL ROADMAP](#) by **May 17, 2026** (9:00 PM GMT+7).

